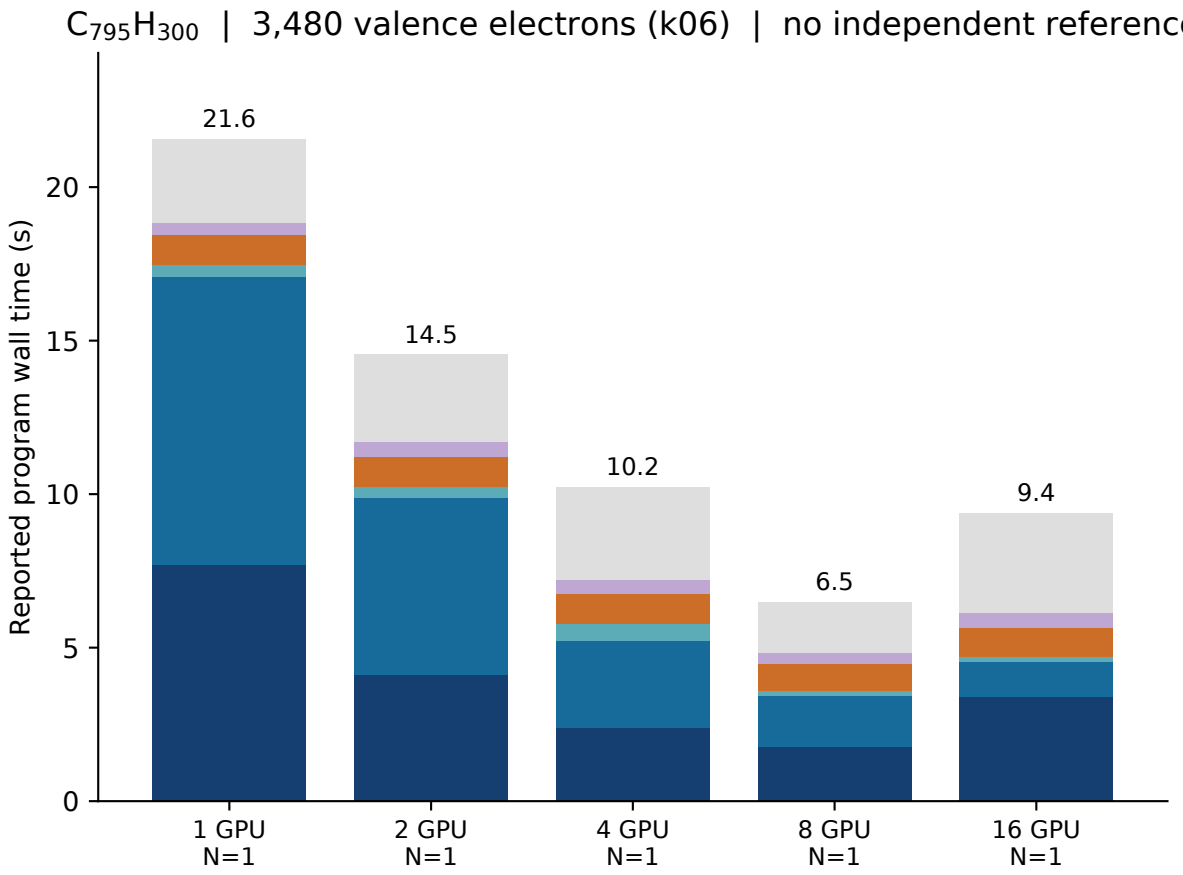
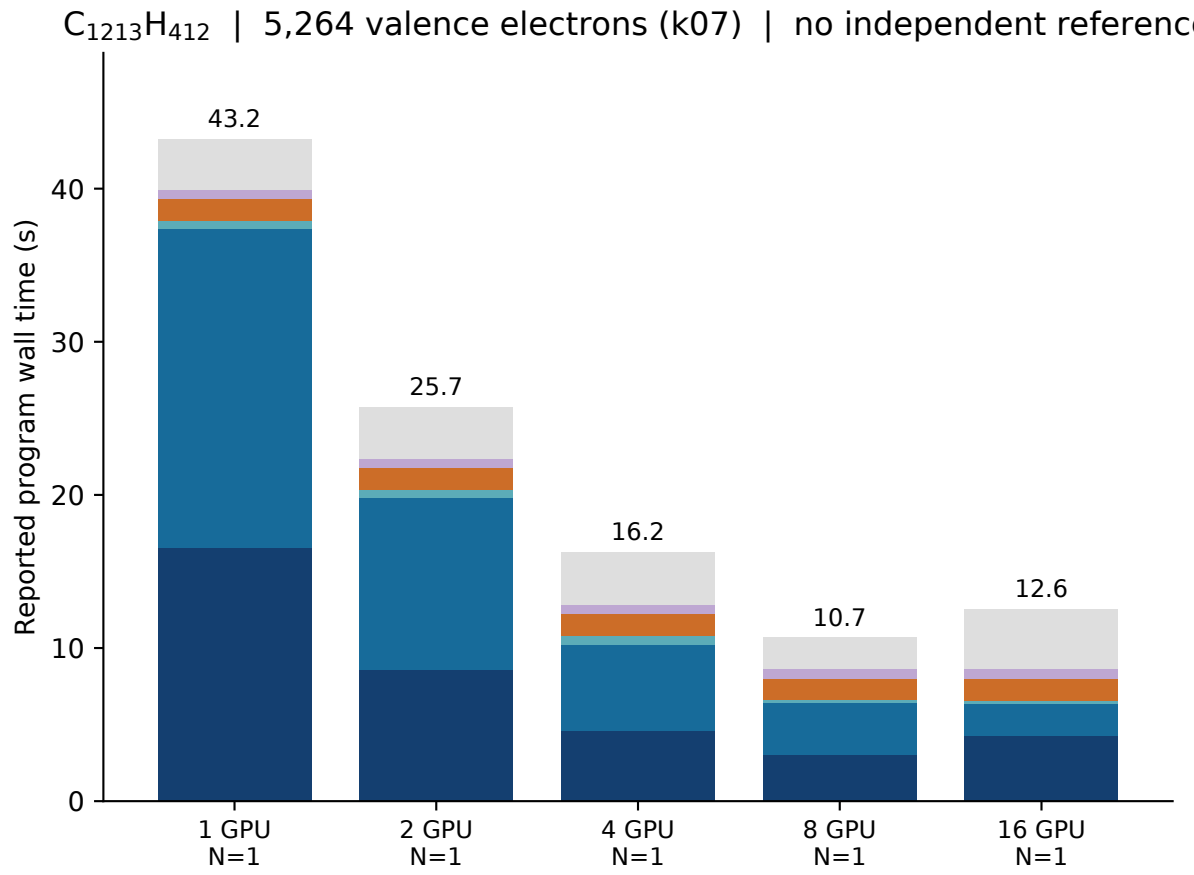
# Complete SCF wall-time breakdown
## Converged, physics-validated runs | all symmetry sectors | FP64

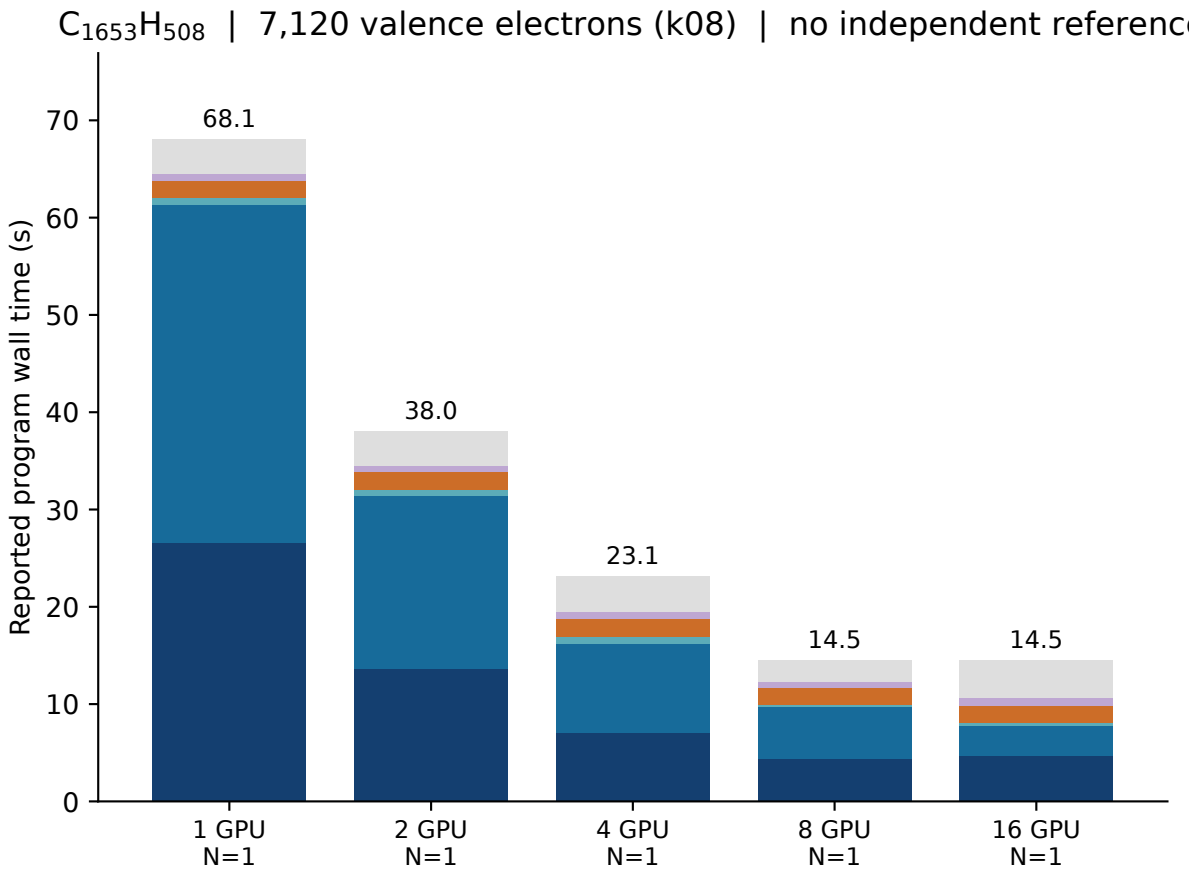
### $C_{795}H_{300}$ | 3,480 valence electrons (k06) | no independent reference



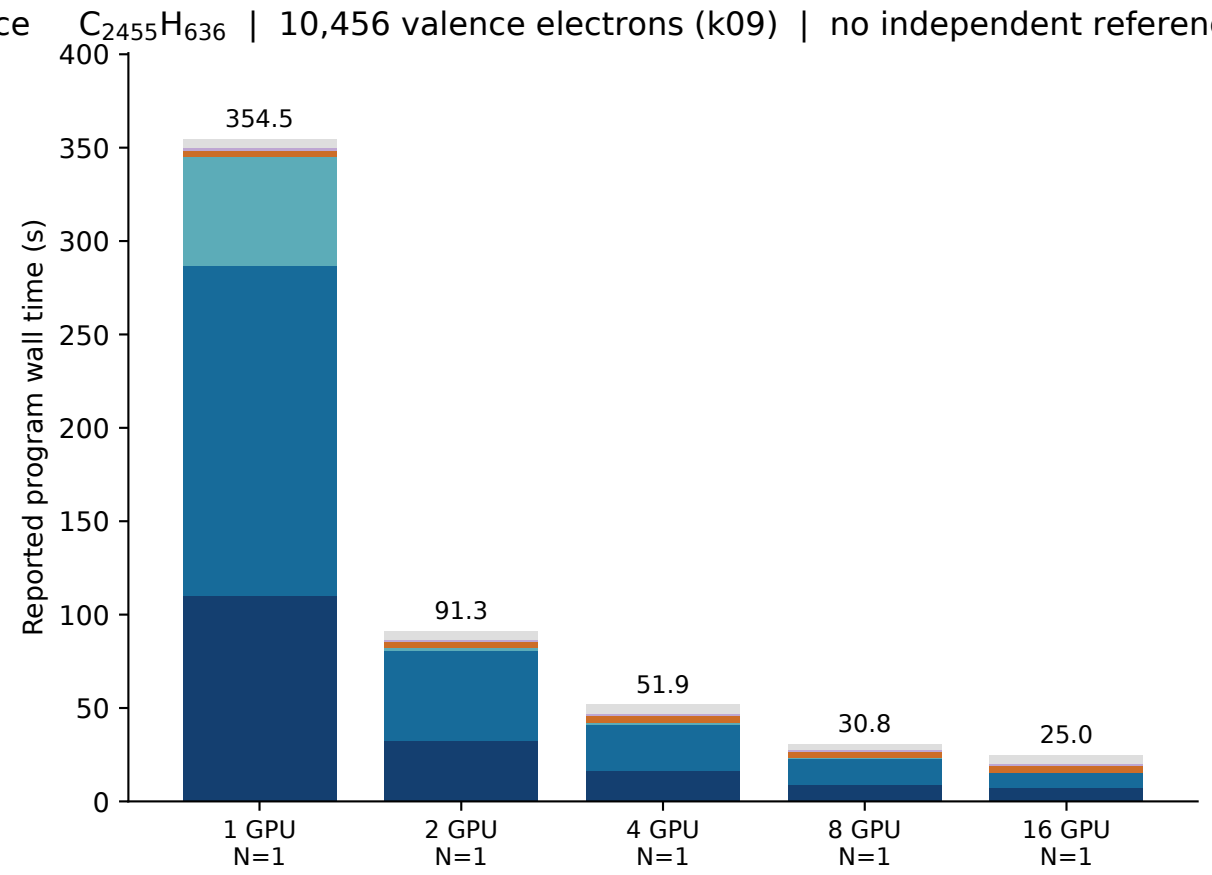

### $C_{1213}H_{412}$ | 5,264 valence electrons (k07) | no independent reference



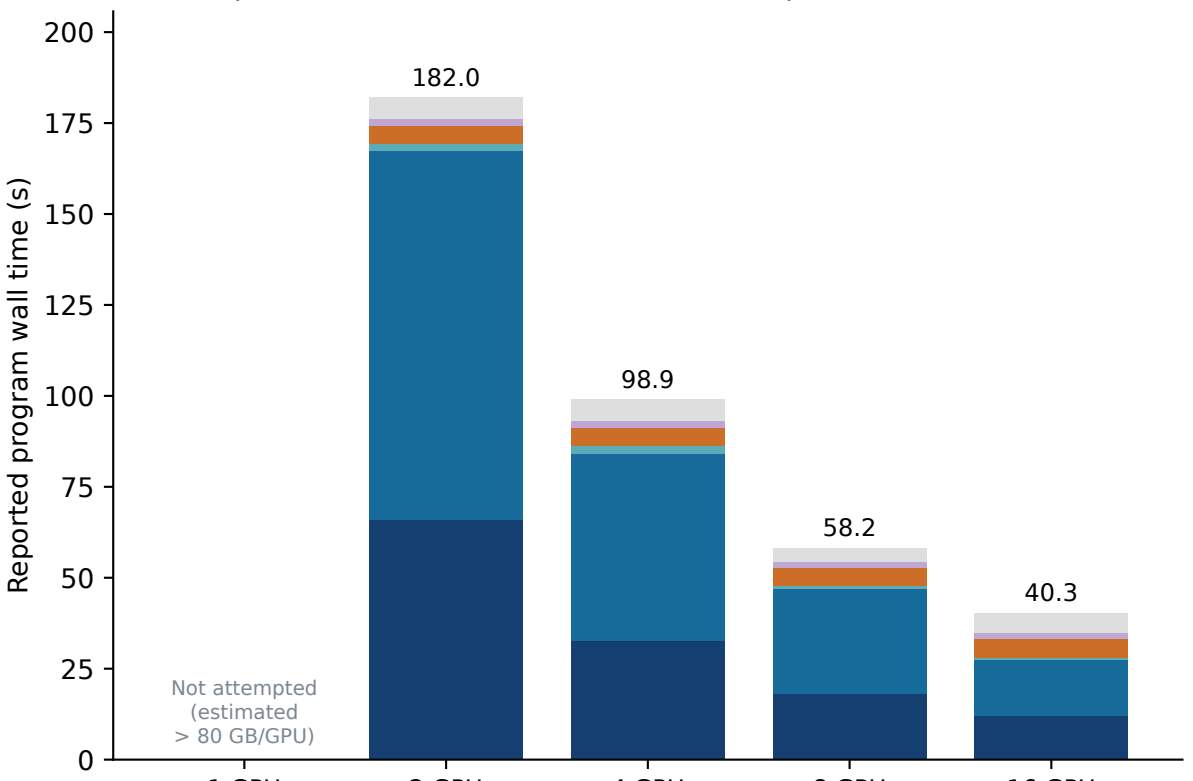

### $C_{1653}H_{508}$ | 7,120 valence electrons (k08) | no independent reference



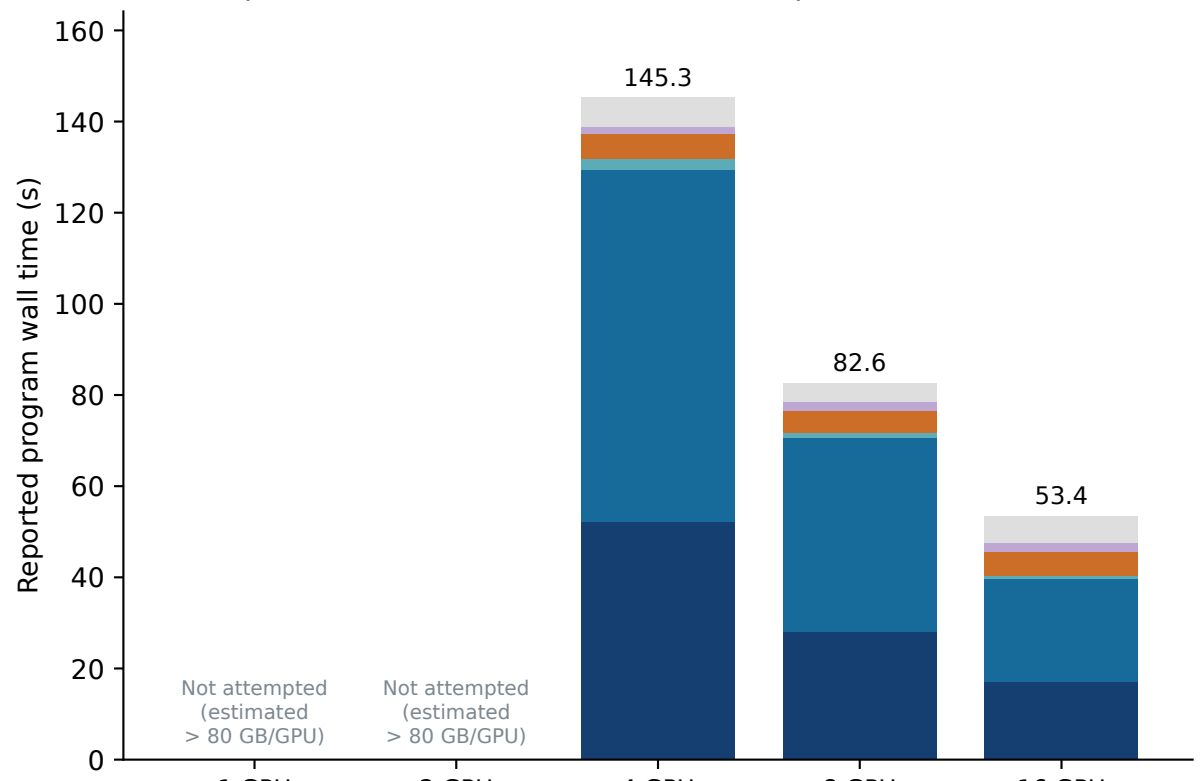

### $C_{2455}H_{636}$ | 10,456 valence electrons (k09) | no independent reference



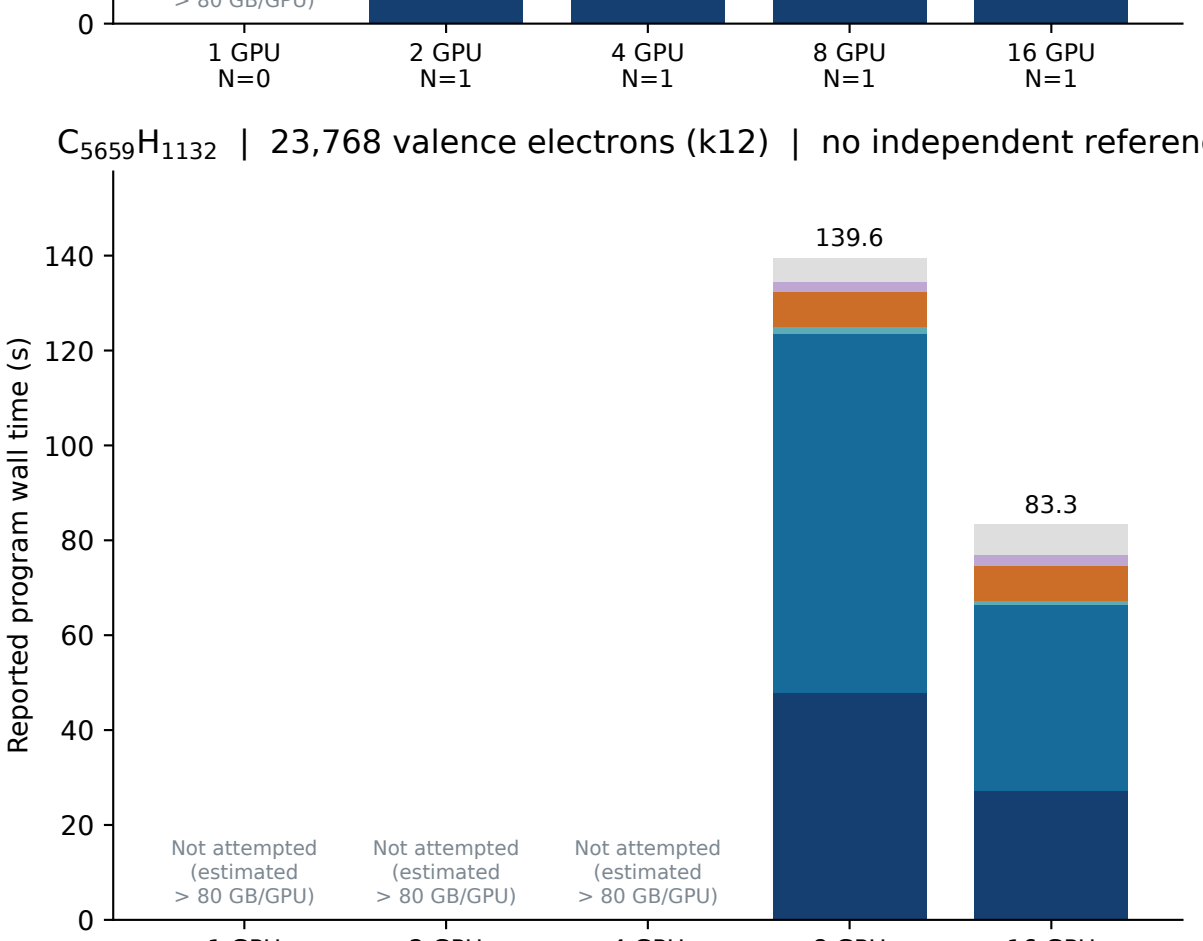

### $C_{3469}H_{804}$ | 14,680 valence electrons (k10) | no independent reference



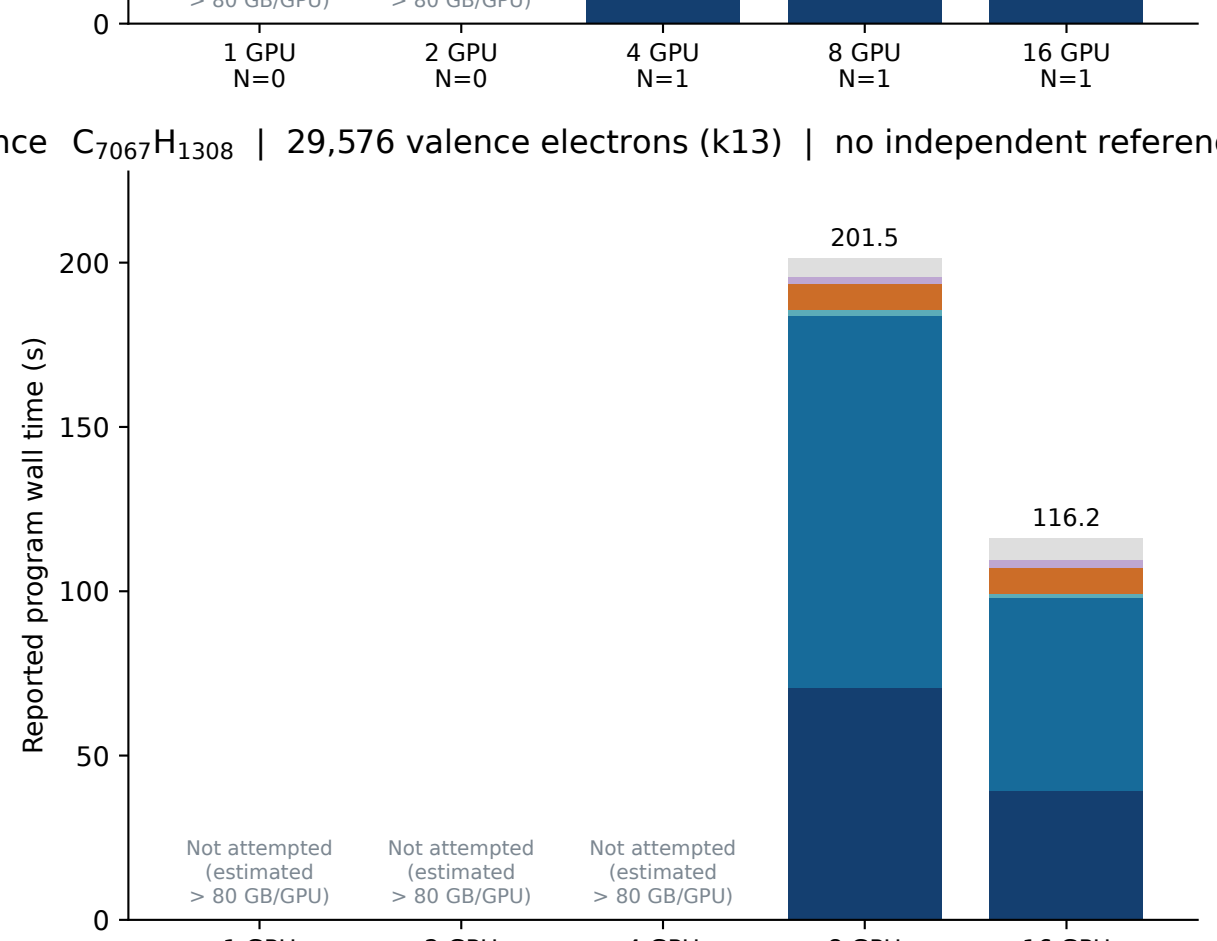

### $C_{4601}H_{988}$ | 19,392 valence electrons (k11) | no independent reference




### $C_{5659}H_{1132}$ | 23,768 valence electrons (k12) | no independent reference




### $C_{7067}H_{1308}$ | 29,576 valence electrons (k13) | no independent reference




### $C_{9449}H_{1572}$ | 39,368 valence electrons (k14) | no independent reference




First eigensolve
Later eigensolves
Density / occupations
Hartree incl. initial solve
Other SCF
Preparation / final reporting

One actual run nearest the median SCF time per configuration; N = validated measured repeats. Missing runs remain blank.
1 MPI rank/node; 16 OpenMP threads/node. CUDA initialization and rank waits included; imports and archive writing excluded.
Host BLAS = 4 threads; small Ritz solve on the GPU. One array per sector where a sector has one GPU; at 8/16 GPUs the sector basis is shared by its GPUs (slabs of at most 1 GiB on 2 GPUs, rounds of 4 GiB on 4).
Same source, physical inputs and runtime controls within each panel. Concurrent GPU event totals are not stacked.
Solver controls FF_MaxIter 4, Chebdav_Degree 30, Chebyshev_Degree 15; master 5fc7363 (2026-10-22), code defaults, double precision, first calculations (no symmetry cache).
C2455H636 on 1 GPU and C3469H804 on 2 GPUs are launched without the storage and allocator pins of the environment file: the code decides where the states live.
Each run compared with the same cell of series F11 (code of 2026-10-21): totals identical to series F11 to the last bit, same SCF steps. Not with a reference archive.